Introduction to Probability and Statistics

Conditional Probability

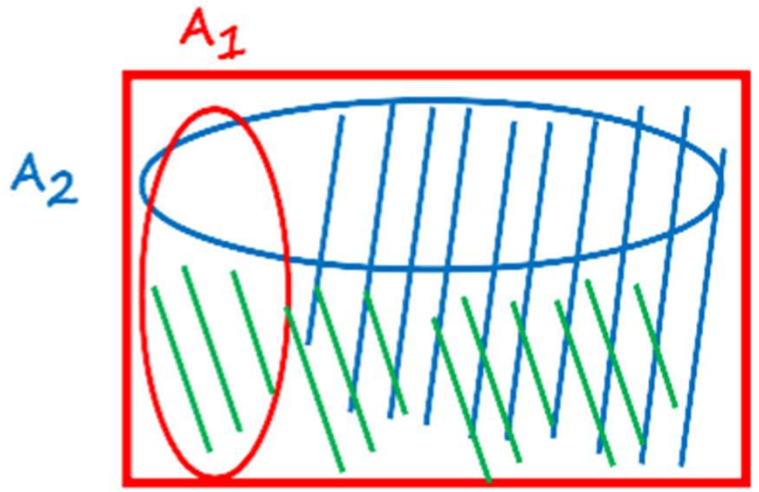
Recall

- Consequences of the Axioms of Probability
  - $\mathbf{P}(A \cup B \cup C)$
  - $\mathbf{P}(\emptyset)$
- A systematic approach to dealing w/ uncertainty + Examples
- Probability Law: Example w/ Countably Infinite Sample Space

Interpreting the union bound and the Bonferroni inequality

$$P(A_1 \cup A_2) \leq P(A_1) + P(A_2)$$

- Suppose that:
  - Very few students are smart A_1
  - Very few students are beautiful A_2
- Then: very few students are smart or beautiful



- Suppose that:
  - Most of the students are smart
  - Most students are beautiful
- Then: most students are smart and beautiful

$$P(A_1 \cap A_2) \geq P(A_1) + P(A_2) - 1$$

The Bonferroni inequality

$$P(A_1 \cap A_2) \geq P(A_1) + P(A_2) - 1$$

$$P(A_1 \cup A_2) \leq P(A_1) + P(A_2)$$

$$P(A_1 \cap A_2)^c = P(A_1^c \cup A_2^c) \leq P(A_1^c) + P(A_2^c)$$

$$1 - P(A_1 \cap A_2) \leq 1 - P(A_1) + 1 - P(A_2)$$

$$P(A_1 \cap \ldots \cap A_n) \geq P(A_1) + \cdots + P(A_n) - (n-1)$$

$$P(A_1 \cap \ldots \cap A_n)^c = P(A_1^c \cup \ldots \cup A_n^c) \leq P(A_1^c) + \ldots + P(A_n^c)$$

$$1 - P(A_1 \cap \ldots \cap A_n) \leq (1 - P(A_1)) + \ldots + (1 - P(A_n))$$

Conditioning & Bayes' rule

- Let's say that you roll two dice and their sum comes out to be S.
  - ❖ If S = 9, what can we say about the first and second roll?
    - ✓ What is the probability that first roll is 1 or 2?

New knowledge should be used to modify our original beliefs.



Revised probabilities are called the conditional probabilities.

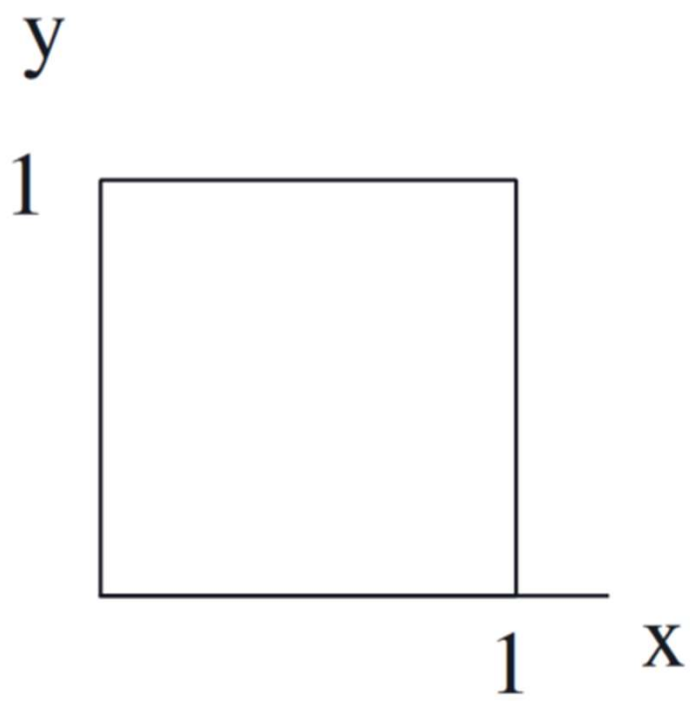
Today's Agenda

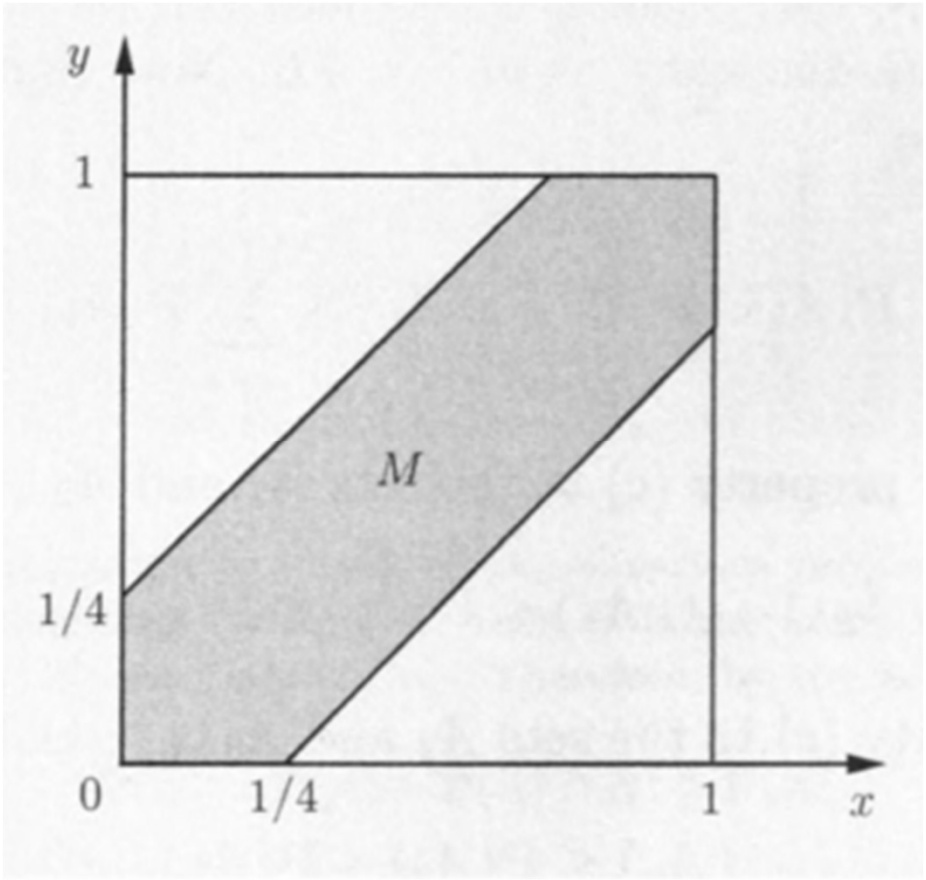
- Romeo & Juliet (example from last class)
- Conditional Probability
  - Intuition
  - Definition
  - Examples

Example 1.5. Romeo and Juliet have a date at a given time, and each will arrive at the meeting place with a delay between 0 and 1 hour, with all pairs of delays being equally likely. The first to arrive will wait for 15 minutes and will leave if the other has not yet arrived. What is the probability that they will meet?

$\Omega = \{(x.y)|\ 0 \leq x \leq 1, 0 \leq y \leq 1\}$

$M = \{(x,y)|\ |x-y| \leq \frac{1}{4}, 0 \leq x \leq 1, 0 \leq y \leq 1\}$

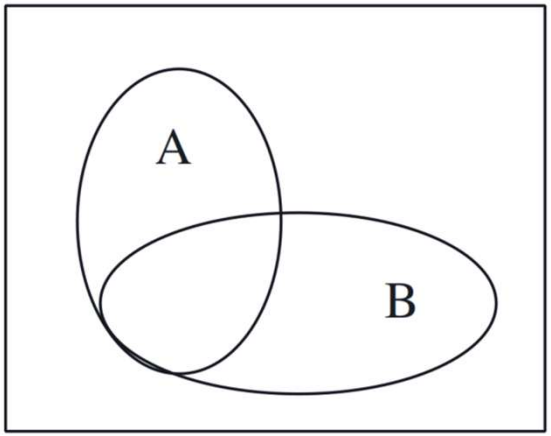



y
1
M
1/4
0
1/4
1
x

Conditional Probability: Intuition

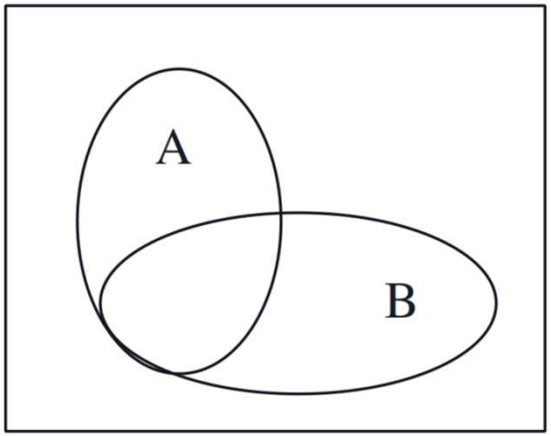
- **Probability Law**:
  - Encodes our “belief” about the likelihood of an event in a random experiment.
- **Conditional Probability Law**:
  - Encodes our “updated belief” about the likelihood of an event **IF** we have “additional partial information” about the results of the random experiment.

Conditional Probability



- $\mathbf{P}(A \mid B) =$ probability of A, given that B occurred
  - B is our new universe

Conditional Probability: Definition



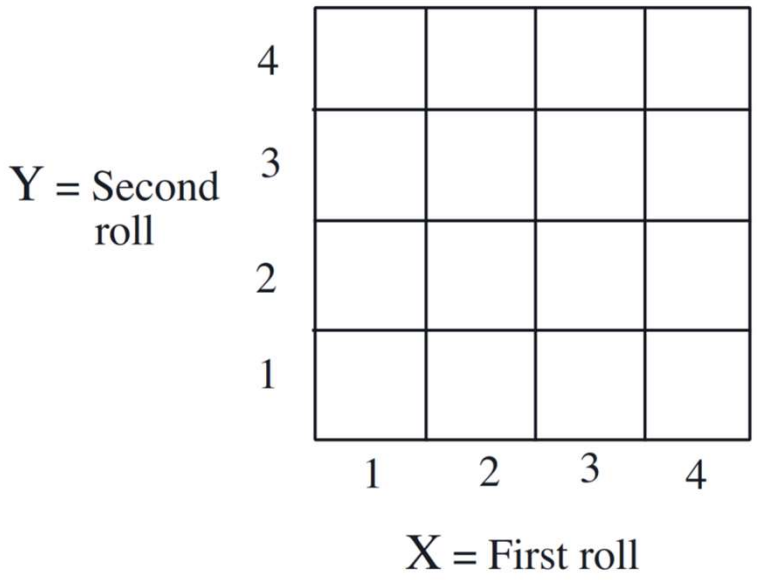
- $\mathbf{P}(A \mid B) =$ probability of A, given that B occurred
  - B is our new universe
- **Definition:** Assuming $\mathbf{P}(B) \neq 0$,

$$\mathbf{P}(A \mid B) = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(B)}$$

$\mathbf{P}(A \mid B)$ is undefined if $\mathbf{P}(B) = 0$

Axioms

Example: Die Roll



- Let B be the event: $\min(X, Y) = 2$
- Let $M = \max(X, Y)$
- $\mathbf{P}(M = 1 \mid B) =$
- $\mathbf{P}(M = 2 \mid B) =$

Example: Successive Coin Tosses

Example 1.6. We toss a fair coin three successive times. We wish to find the conditional probability $\mathbf{P}(A \mid B)$ when A and B are the events

$$A = \{\text{more heads than tails come up}\}, \qquad B = \{\text{1st toss is a head}\}.$$

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The sample space consists of eight sequences.

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}.$$

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Example 1.6. We toss a fair coin three successive times. We wish to find the conditional probability $\mathbf{P}(A \mid B)$ when A and B are the events

$$A = \{\text{more heads than tails come up}\}, \qquad B = \{\text{1st toss is a head}\}.$$

The sample space consists of eight sequences.

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\},$$

which we assume to be equally likely. The event B consists of the four elements HHH, HHT, HTH, HTT, so its probability is

$$\mathbf{P}(B) = \frac{4}{8}.$$

The event $A \cap B$ consists of the three elements HHH, HHT, HTH, so its probability is

$$\mathbf{P}(A \cap B) = \frac{3}{8}.$$

Thus, the conditional probability $\mathbf{P}(A \mid B)$ is

$$\mathbf{P}(A \mid B) = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(B)} = \frac{3/8}{4/8} = \frac{3}{4}.$$

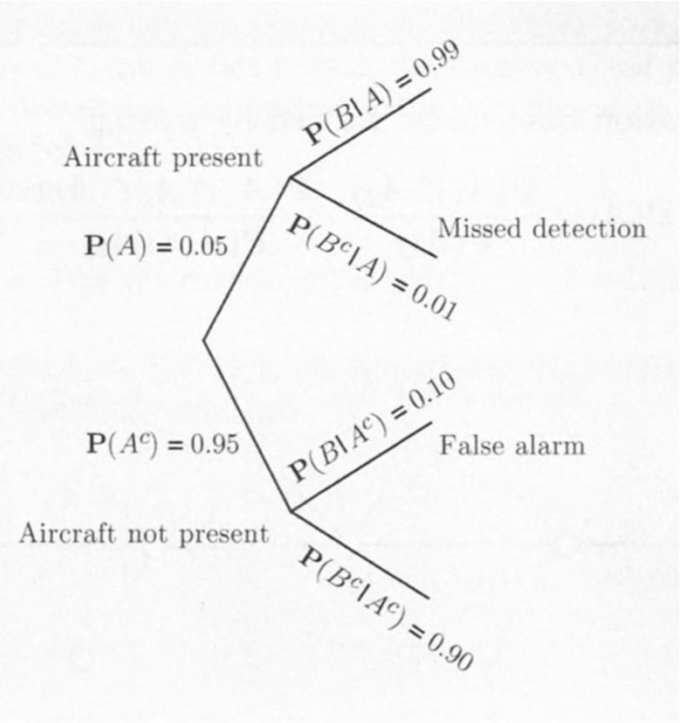
Example: Flying Under the Radar

Example 1.9. Radar Detection. If an aircraft is present in a certain area, a radar detects it and generates an alarm signal with probability 0.99. If an aircraft is not present. the radar generates a (false) alarm, with probability 0.10. We assume that an aircraft is present with probability 0.05. What is the probability of no aircraft presence and a false alarm? What is the probability of aircraft presence and no detection?

Example: Flying Under the Radar

Event A: Airplane is flying above

Event B: Something registers on radar screen



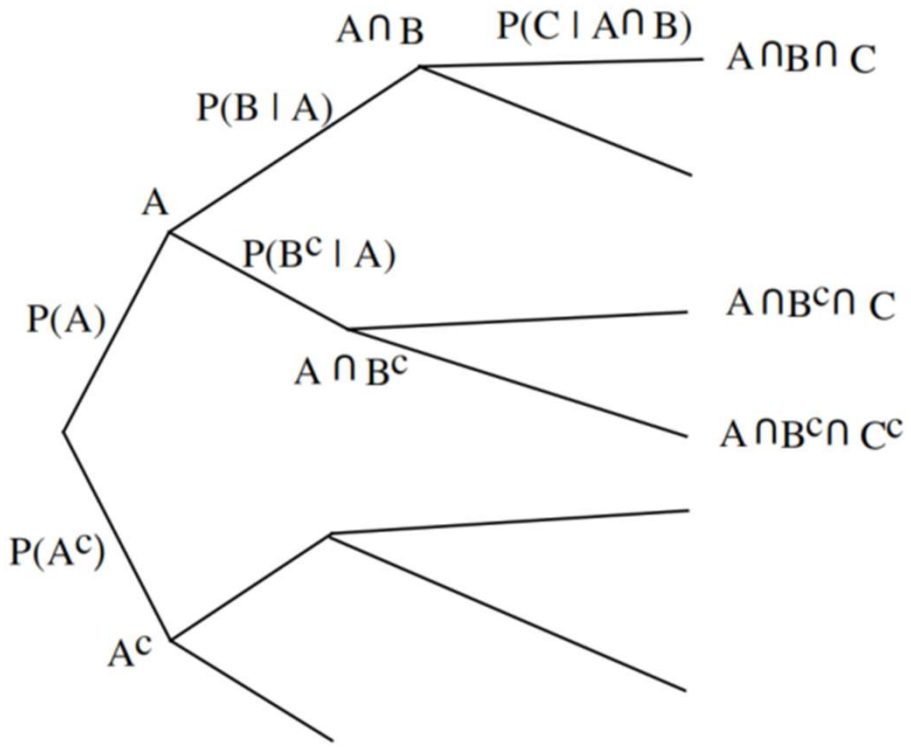
$P(A \cap B) =$

$P(B) =$

$P(A \mid B) =$

Multiplication Rule

$$\mathbf{P}(A \cap B \cap C) = \mathbf{P}(A) \,.\, \mathbf{P}(\, B \mid A \,) \,.\, \mathbf{P}(C \mid A \cap B)$$



Multiplication Rule

Multiplication Rule

Assuming that all of the conditioning events have positive probability, we have

$$\mathbf{P}\left(\cap_{i=1}^{n} A_i\right) = \mathbf{P}(A_1)\mathbf{P}(A_2 \mid A_1)\mathbf{P}(A_3 \mid A_1 \cap A_2)\cdots\mathbf{P}\left(A_n \mid \cap_{i=1}^{n-1} A_i\right).$$